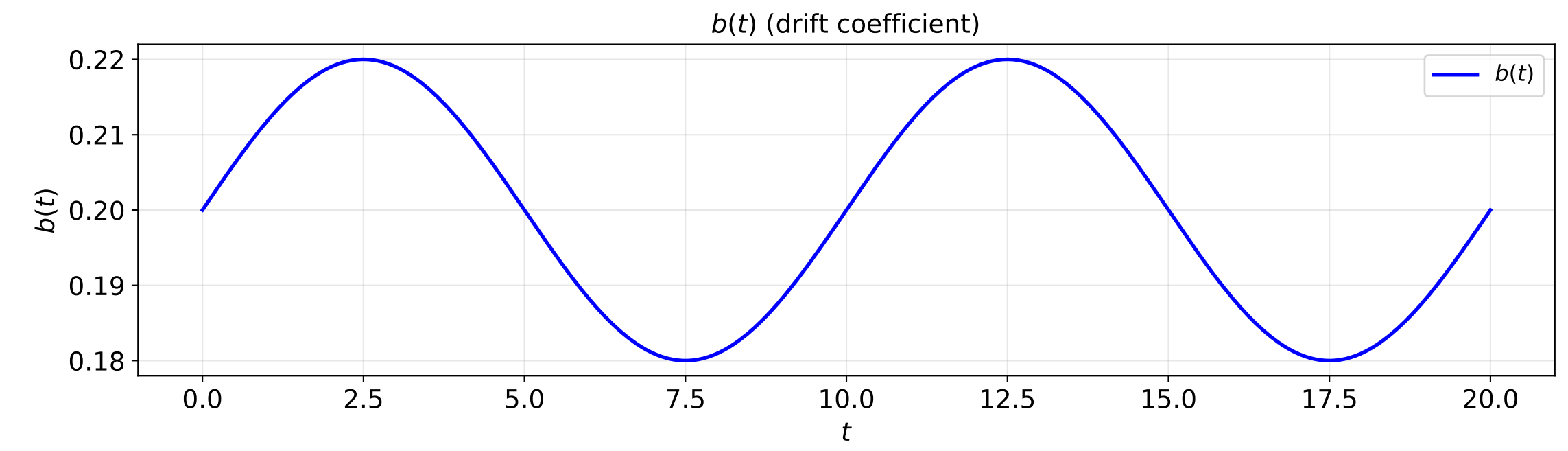
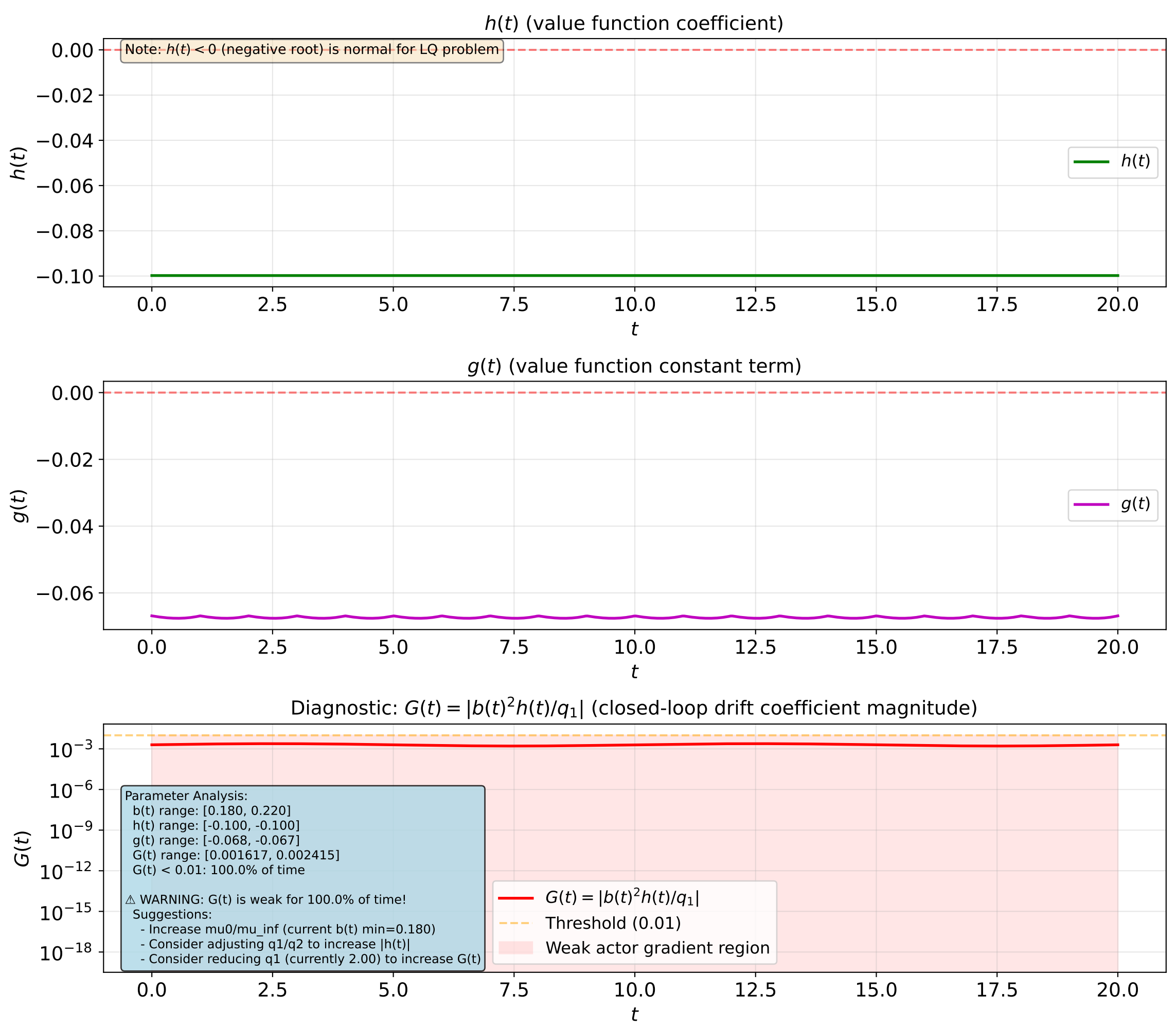

$b(t)$ (drift coefficient)
$b(t)$
t
$h(t)$ (value function coefficient)
Note: $h(t) < 0$ (negative root) is normal for LQ problem
$h(t)$
t


$g(t)$ (value function constant term)
$g(t)$
t
Diagnostic: $G(t) = |b(t)^2 h(t)/q_1|$ (closed-loop drift coefficient magnitude)
Parameter Analysis:
b(t) range: [0.180, 0.220]
h(t) range: [-0.100, -0.100]
g(t) range: [-0.068, -0.067]
G(t) range: [0.001617, 0.002415]
G(t) < 0.01: 100.0% of time
⚠ WARNING: G(t) is weak for 100.0% of time!
Suggestions:
- Increase mu0/mu_inf (current b(t) min=0.180)
- Consider adjusting q1/q2 to increase |h(t)|
- Consider reducing q1 (currently 2.00) to increase G(t)
$G(t) = |b(t)^2 h(t)/q_1|$
Threshold (0.01)
Weak actor gradient region
$G(t)$
t